

DBAD 4000 – Advanced Database
Project Phase 1: Database Modelling
DriveEase Car Rental Management Database
Technical Report

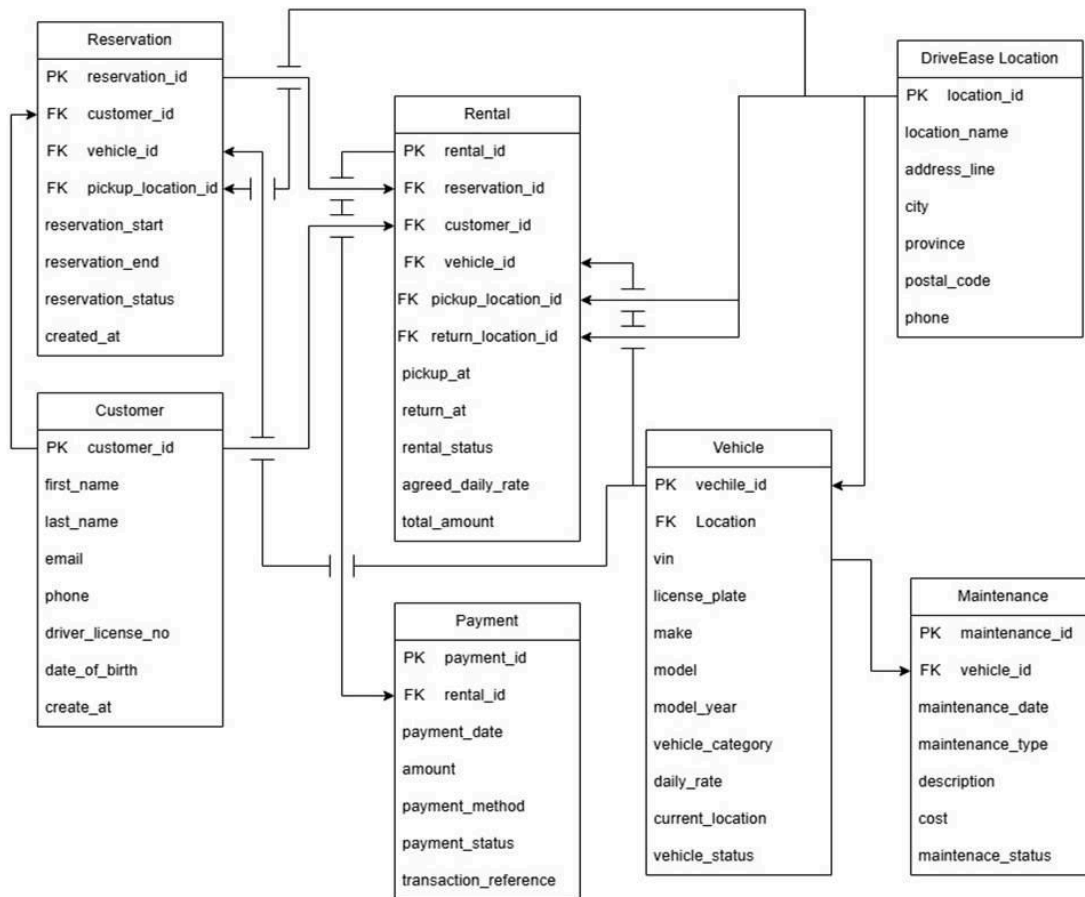
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Course: DBAD 4000 – Advanced Database

Phase: 1 – Database Modelling

EER Diagram

The EER diagram for the DriveEase rental database is shown below. The diagram was created using Draw.io to represent the main entities, their attributes, primary keys, foreign keys, and relationships within the system.



The database consists of seven main entities: CUSTOMER, LOCATION, VEHICLE, RESERVATION, RENTAL, PAYMENT, and MAINTENANCE. The relationships between these entities allow the system to track customers, vehicles, rental locations, reservations, completed rentals, payments, and vehicle maintenance history.

Relational Schema:

- ® CUSTOMER (customer_id PK, first_name, last_name, email, phone, driver_license_no, date_of_birth, created_at).
- ® LOCATION (location_id PK, location_name, address_line, city, province, postal_code, phone).
- ® VEHICLE (vehicle_id PK, vin, license_plate, make, model, model_year, vehicle_category, daily_rate, current_location_id FK → LOCATION, vehicle_status).

- ® RESERVATION (reservation_id PK, customer_id FK → CUSTOMER, vehicle_id FK → VEHICLE, pickup_location_id FK → LOCATION, reservation_start, reservation_end, reservation_status, created_at).
- ® RENTAL (rental_id PK, reservation_id FK → RESERVATION, customer_id FK → CUSTOMER, vehicle_id FK → VEHICLE, pickup_location_id FK → LOCATION, return_location_id FK → LOCATION, pickup_at, return_at, rental_status, agreed_daily_rate, total_amount).
- ® PAYMENT (payment_id PK, rental_id FK → RENTAL, payment_date, amount, payment_method, payment_status, transaction_reference).
- ® MAINTENANCE (maintenance_id PK, vehicle_id FK → VEHICLE, maintenance_date, maintenance_type, description, cost, maintenance_status).

Referential Integrity Rules:

Foreign Key	Referenced Key	Delete / Update Rule	Reason
VEHICLE.current_location_id	LOCATION.location_id	ON DELETE RESTRICT & ON UPDATE CASCADE	A location should not be deleted while vehicles reference it.
RESERVATION.customer_id	CUSTOMER.customer_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves reservation history.
RESERVATION.vehicle_id	VEHICLE.vehicle_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves reservation history.

RESERVATION.pickup_location_id	LOCATION.location_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves location history.
RENTAL.reservation_id	RESERVATION.reservation_id	ON DELETE RESTRICT & ON UPDATE CASCADE	A rental should retain its reservation link.
RENTAL.customer_id	CUSTOMER.customer_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves rental history.
RENTAL.vehicle_id	VEHICLE.vehicle_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves vehicle rental history.
RENTAL.pickup_location_id	LOCATION.location_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves pickup history.
RENTAL.return_location_id	LOCATION.location_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves return history.

PAYMENT.rental_id	RENTAL.rental_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Payments must remain linked to their rental.
MAINTENANCE.vehicle_id	VEHICLE.vehicle_id	ON DELETE RESTRICT & ON UPDATE CASCADE	Preserves maintenance history.

Normalization to 3NF:

4.3.1 Functional Dependencies:

- ® CUSTOMER: customer_id → first_name,last_name, email, phone, driver_license_no,date_of_birth,created_at.
- ® LOCATION: location_id → location_name,address_line, city, province,postal_code, phone.
- ® VEHICLE: vehicle_id → vin,license_plate, make, model,model_year, vehicle_category, daily_rate, current_location_id, vehicle_status.
- ® RESERVATION: reservation_id → customer_id, vehicle_id,pickup_location_id, reservation_start, reservation_end, reservation_status, created_at.
- ® RENTAL: rental_id → reservation_id, customer_id, vehicle_id, pickup_location_id, return_location_id,pickup_at,return_at,rental_status,agreed_daily_rate,total_amount.
- ® PAYMENT: payment_id → rental_id,payment_date,amount,payment_method, payment_status, transaction_reference.
- ® MAINTENANCE: maintenance_id → vehicle_id,maintenance_date, maintenance_type, description, cost, maintenance_status.

4.3.2 1NF: The design is in First Normal Form because each attribute contains a single atomic value. For example, a customer has a separate first name, last name, email, and phone attributes. Multiple payments are stored as separate Payment rows rather than as repeated payment columns in Rental.

4.3.3 2NF: The design is in Second Normal Form because the tables use single-column primary keys. Therefore, no non-key attribute can depend on only part of a composite primary key. Each non-key attribute depends on the complete primary key of its table.

4.3.4 3NF: The design is in Third Normal Form because non-key attributes describe the entity identified by the primary key and are not stored through unnecessary transitive dependencies. For example, location details are stored in LOCATION rather than repeated in VEHICLE, RESERVATION, and RENTAL. Vehicle details are stored in VEHICLE rather than repeated in RENTAL.

This separation reduces duplication and makes future updates more reliable. It also gives later phases a clean operational source for generating synthetic high-volume data and loading a separate analytical star schema.

Data Dictionary:

Table	Attribute	Description	PostgreS QL Type	Key	Validati on	Sample / Default
CUSTOMER	customer_id	Unique customer identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
CUSTOMER	first_name	Customer first name	varchar(50)	—	NOT NULL	Alex
CUSTOMER	last_name	Customer last name	varchar(50)	—	NOT NULL	Smith
CUSTOMER	email	Customer email address	varchar(254)	—	NOT NULL; UNIQUE; valid email format	Alex@gmail.com
CUSTOMER	phone	Customer phone number	varchar(20)	—	NOT NULL	403-555-0100
CUSTOMER	driver_license_no	Unique driver licence number	varchar(30)	—	NOT NULL; UNIQUE	AB123456
CUSTOMER	date_of_birth	Customer date of birth	date	—	NOT NULL; reasonable adult/rental	1995-06-15

					validation	
CUSTOMER	created_at	Date/time customer record was created	timestamp	—	NOT NULL	Current timestamp
LOCATION	location_id	Unique rental location identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
LOCATION	location_name	Branch name	varchar(100)	—	NOT NULL	Downtown Calgary
LOCATION	address_line	Street address	varchar(150)	—	NOT NULL	100 Main Street
LOCATION	city	City	varchar(80)	—	NOT NULL	Calgary
LOCATION	province	Canadian province	varchar(50)	—	NOT NULL	Alberta
LOCATION	postal_code	Canadian postal code	varchar(7)	—	NOT NULL; format validation	T2P 1A1
LOCATION	phone	Branch phone number	varchar(20)	—	NOT NULL	403-888-2222
VEHICLE	vehicle_id	Unique vehicle identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
VEHICLE	vin	Vehicle identification number	varchar(17)	—	NOT NULL; UNIQUE	1HGCM826
VEHICLE	license_plate	Vehicle plate number	varchar(15)	—	NOT NULL; UNIQUE	ABC123
VEHICLE	make	Vehicle manufacturer	varchar(50)	—	NOT NULL	Toyota
VEHICLE	model	Vehicle model	varchar(50)	—	NOT NULL	Corolla

VEHICLE	model_year	Vehicle model year	smallint	—	NOT NULL; reasonable year range	2025
VEHICLE	vehicle_category	Rental category	varchar(30)	—	NOT NULL; controlled values	SUV
VEHICLE	daily_rate	Standard daily rental rate	numeric(10,2)	—	NOT NULL; >= 0	89.99
VEHICLE	current_location_id	Vehicle's current branch	integer	FK → LOCATION	NOT NULL	1
VEHICLE	vehicle_status	Current vehicle status	varchar(20)	—	NOT NULL; controlled values	Available
RESERVATION	reservation_id	Unique reservation identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
RESERVATION	customer_id	Customer making reservation	integer	FK → CUSTOMER	NOT NULL	1
RESERVATION	vehicle_id	Vehicle reserved	integer	FK → VEHICLE	NOT NULL	10
RESERVATION	pickup_location_id	Planned pickup branch	integer	FK → LOCATION	NOT NULL	1
RESERVATION	reservation_start	Planned rental start	timestamp	—	NOT NULL	2026-10-10 09:00
RESERVATION	reservation_end	Planned rental end	timestamp	—	NOT NULL; after start	2026-10-13 09:00
RESERVATION	reservation_status	Current reservation status	varchar(20)	—	NOT NULL; controlled values	Confirmed

RESERVATION	created_at	Reservation creation time	timestamp tz	—	NOT NULL	Current timestamp
RENTAL	rental_id	Unique rental transaction identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
RENTAL	reservation_id	Source reservation	integer	FK → RESERVATION	NOT NULL	25
RENTAL	customer_id	Customer renting vehicle	integer	FK → CUSTOMER	NOT NULL	1
RENTAL	vehicle_id	Vehicle rented	integer	FK → VEHICLE	NOT NULL	10
RENTAL	pickup_location_id	Actual pickup branch	integer	FK → LOCATION	NOT NULL	1
RENTAL	return_location_id	Actual return branch	integer	FK → LOCATION	NOT NULL	2
RENTAL	pickup_at	Actual pickup date/time	timestamp tz	—	NOT NULL	2026-10-10 09:10
RENTAL	return_at	Actual return date/time	timestamp tz	—	NULL until returned; after pickup	2026-10-13 10:00
RENTAL	rental_status	Current rental status	varchar(20)	—	NOT NULL; controlled values	Completed
RENTAL	agreed_daily_rate	Rate agreed for rental	numeric(10,2)	—	NOT NULL; >= 0	89.99
RENTAL	total_amount	Total rental charge	numeric(12,2)	—	NOT NULL; >= 0	269.97
PAYMENT	payment_id	Unique payment identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated

PAYMENT	rental_id	Rental paid by transaction	integer	FK → RENTAL	NOT NULL	100
PAYMENT	payment_date	Payment date/time	timestampz	—	NOT NULL	2026-10-13 10:15
PAYMENT	amount	Payment amount	numeric(12,2)	—	NOT NULL; > 0	269.97
PAYMENT	payment_method	Payment method	varchar(20)	—	NOT NULL; controlled values	Credit Card
PAYMENT	payment_status	Payment processing status	varchar(20)	—	NOT NULL; controlled values	Completed
PAYMENT	transaction_reference	External/payment reference	varchar(80)	—	UNIQUE when present	TXN-10001
MAINTENANCE	maintenance_id	Unique maintenance identifier	integer	PK; generated identity	NOT NULL; UNIQUE	Generated
MAINTENANCE	vehicle_id	Vehicle receiving maintenance	integer	FK → VEHICLE	NOT NULL	10
MAINTENANCE	maintenance_date	Date maintenance occurred	date	—	NOT NULL	2026-09-20
MAINTENANCE	maintenance_type	Type of maintenance	varchar(50)	—	NOT NULL	Oil Change
MAINTENANCE	description	Maintenance details	varchar(500)	—	NULL allowed	Routine service
MAINTENANCE	cost	Maintenance cost	numeric(10,2)	—	NOT NULL; >= 0	125.50
MAINTENANCE	maintenance_status	Maintenance status	varchar(20)	—	NOT NULL; controlled values	Completed

Major Design Decisions

The DriveEase database was designed to focus on the core vehicle rental business using synthetic data. Vehicles have unique VINs and license plates and are assigned to one current location. Reservations and actual rentals are stored separately, while pickup and return locations are tracked independently. Payments and maintenance are also separated to support multiple transactions and maintenance records per rental or vehicle. The design also preserves historical records for reporting and auditing, while vehicle availability and overlapping rental rules will be handled through business rules in later phases.

Individual Contribution Table

Team member	Responsibilities	Specific Contribution to this Phase
<i>Vivek Patel</i>	Created the initial relational schema and data dictionary	Worked on relational schema and data dictionary. Made many design choices
<i>Saihaj Mann</i>	Create an EER diagram.	Created EER diagram.
Kyle Ince	Technical Sheet; EER diagram review;	Did the write up of the technical sheet; reviewed the EER diagram
<i>Nathan Woyessa</i>	Github repository; read me; submission;	Managed the repo, created the read me and finalized the project files for submission

